

Review

Mad honey intoxication: A systematic review on the 1199 cases

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ABSTRACT

Mad honey, produced by honeybees from the nectars of *Rhododendron* genus (*R. ponticum* and *R. luteum*) flowers, is widely used in indigenous medicine, especially in the treatment of hypertension and sexual dysfunction. However, the consumption of this honey can result in intoxication soon after. The diagnosis of honey poisoning and a full understanding of its treatment is important for both effective and immediate treatment, and also for the prevention of unnecessary costs. Upon the evaluation of approximately 34 years of case reports between 1981 and 2014, it was found that the cases of poisoning were more frequently reported in males (75.17%) and between the ages 41 to 65. The most common complaints related to honey poisoning were dizziness, nausea, presyncope and the ECG findings were: sinus bradycardia (79.58%), complete atrioventricular block (45.83%), atrioventricular block (30.91%), ST-segment elevation (22.63%), and nodal rhythm (11.27%). As a result of the evaluation of 1199 cases, it was found that no deaths were reported. The patients were most frequently treated with 0.5 mg atropine (37.79%), 1 mg atropine (49.73%), saline (iv fluid) (65.35%), and generally the patients were discharged within 24 h after recovery.

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1. Introduction

The *Rhododendron* genus, which is among the Ericaceae family, is one of the largest genera of vascular plants covering a large part of the northern hemisphere (Popescu and Kopp, 2013). *R. ponticum* and *R. luteum* plants which are the sources for rhododendron (or mad) honey grow mainly in the Black Sea region of Turkey, Japan,

Nepal, Brazil, Europe and some parts of North America (Davis, 1978). In this genus, 208 compounds have been isolated, composed of mostly flavonoids and diterpenoids (Quiang et al., 2011). Most of those diterpenoids are grayane-type diterpenoids, polyhydroxylated cyclic hydrocarbons that do not contain nitrogen (Lampe, 1988). It is reported that mad honey intoxication is largely associated with lipid-soluble grayanotoxins similar to the alkaloids veratridine, acotinine and batrachotoxin (Cestele and Catterall, 2000). GTXs tend to bind to the activated state of sodium channels and cause persistent activation at resting membrane potential. They lead to blockage of sodium channel inactivation and shift of

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the voltage dependence of activation to more negative potentials (Narahashi and Seyama, 1974; Catterall, 1980).

Historically, the phenomenon of honey intoxication is known from ancient times. Xenophon (B.C. 434–354) mentions an event of intoxication of 10,000 soldiers during their stay at the coast of the Black Sea with the Greek Army (Lampe, 1988). The first report on the medicinal properties of the mad honey can be found in Aristotle who declares “at Trapezus honey from boxwood has a heavy scent, and they say that healthy men go mad, but that epileptics are cured by immediately” (Aristotle, 1936). Dioscorides (40–90 AD) referring to mad honey “rubbed on with costum it heals sunburn, and with salt it takes away bruises” (Dioscorides, 2000). Pliny the Elder (23–79 AD) says that mad honey “taken in wine is a remedy for indispositions caused by eating fish” and “there is nothing better than this honey for softening the skin of females, or combined with aloes, for the treatment of bruises” (Pliny, 1855). Honey was called at that time in Greek as “maenomeno”, “miraculum mellisa” or “wondrous honey” (Halliday, 1922). In the Black Sea region of Turkey, Rhododendron honey is available in the local markets in raw form without any processing and known as “deli bal” or “mad honey” because of its intoxication property, and also as “bitter honey” because of its slightly sharp taste.

The scientific and medical community's primary interest has been the study of the causes of this phenomenon. Subsequently, in the majority of previous studies, nutritional components, the phenolic content, amino acid content, and biochemical structure of Rhododendron honey were already determined (Silici et al., 2008, 2010, 2013, 2014; Silici, 2010; Silici and Karaman, 2014). Generally, the mad honey intoxication case reports are from Trabzon in Turkish Black Sea Region and it is reported by the intoxication patients that the honey is used in the region against peptic ulcer, gastritis, abdominal pain, indigestion, hypertension, flu, arthritis and also for its sexual stimulant and anti-glycemic effects (Bostan et al., 2010; Dilber et al., 2002; Aliyev et al., 2009; Demircan et al., 2009; Gunduz et al., 2008). Furthermore, various researches on the chemical properties of that honey and its useful biological activities are available. Beside this, antimicrobial, antiradical, antioxidant and bioactive properties such as anti-diabetic activity have been reported in various studies as well (Demircan et al., 2009; Ertürk et al., 2009; Öztasan et al., 2005; Silici et al., 2010). Rhododendron honey has been used in the indigenous medicine for a long time particularly as a sexual stimulant and against high blood pressure and high blood glucose levels. Given the large therapeutic use of this honey in the country, intoxication with complaints such as dizziness, vomiting, nausea, weakness and blurred vision is not an uncommon problem in Turkey where most of the reports by some authors (Gunduz et al., 2008; Bostan et al., 2010).

Consequently, mad honey ingestion can cause serious impairments that often require emergency-room care and can be also mistaken for other, more serious traumas. It has been reported in some studies that some patients had been transferred to the intensive care unit (Ergun et al., 2005; Hanci et al., 2010) and that resulted in an increase in the cost of the management. Patients presented to the emergency room exhibiting such symptoms are usually treated with atropine and discharged within one day (Gunduz et al., 2006, 2009). However, there is limited information about the management of the disease and every hospital has its own treatment and follow-up protocol. In the current review, the main focus is to provide an overall review and evaluation of clinical cases reported on honey poisoning with the objective of drawing general observations and conclusions that can help effective treatment and management of such cases.

2. Materials and methods

We searched the official home pages of PubMed, Scopus, Web of Science and Google Scholar to identify information specific to the cases on mad honey. Sex, age, complaints, diastolic and systolic blood pressure (mmHg), heart rate, electrocardiography (ECG) characteristics, amount of honey consumed, usage of another drugs, treatment, recovering time of patients were noted and tabulated (Table 1). SPSS (Statistical Package for Social Science, IBM, Armonk, NY) 15.0 for Windows was used for statistical analysis of the data obtained. Data are shown as mean $\pm$ SD, and data were obtained quantitatively as numbers (%).

3. Results

In the current review, the condition of 1199 cases, which were reported in 84 published articles, were evaluated. They were letter to editor/to the editor (9 papers), case/brief report (52 papers), original contribution (1 paper), short communication (1 paper), clinical study/research (4 papers), original research (10 articles) and published oral/poster presentation (7 presentations) types of publications. The first case report was published by Kerkvliet in 1981. It is a brief case description of four mountaineers that experienced mad honey intoxication. While the first case report was published by Kerkvliet (1981), among these cases, 7 were published before the year 2000, 7 were published between 2000 and 2005, 28 were published between 2005 and 2010 and 42 were published between 2010 and 2015 (Fig. 1). The number of case studies on mad honey reached the highest with 32 articles in 2012 in the open literature. Among these, 16 of them were published in Turkish while one was in German one was French and two were Korean (with English abstracts). Among all the reports 3 were published just in Turkish, three were German and 57 were published just in English. Moreover, 69 of those reports were prepared by Turkish researchers.

Out of 1199 cases reviewed, 890 were male (75.17%) and 294 (24.83%) were female (Table 2). Both systolic (SBP) and diastolic (DBP) blood pressure of patients showed hypotensive tendencies. In the case of DBP, two of them were 20 mmHg, another two were above 80 mmHg, and the rest 991 patients (99.59%) had DBP between 20 and 80 mmHg. In the case of SBP, 312 patients (31.36%) had values below 80 mmHg, 59 patients (5.92%) had values above 120 mmHg, and the rest 626 patients (62.54%) had SBP between 80 and 120 mmHg (Table 2).

Out of all the cases reviewed, heart rates of 1098 of them were achieved. The mean heart rate among these patients were 47.11 ± 15.87 , with the values recorded as low as 28 and as high as 105. Two patients had heart rates below 30 beats/minutes and eight patients had heart rates of 60 beats/minutes and above. The heart rates of 1088 patients (99.08%) were between 30 and 60 beats/minutes.

The complaints of patients were varied and 40 different complaints included (Table 3). Among these complaints dizziness (51.58%), bradycardia (49.92%), nausea (37.58%), vomiting (35.50%), presyncope (27.33%), blurred vision (20.42%) were the most common, followed by hypotension, syncope, fainting and others (Table 3).

The most common ECG pattern were Sinus Bradycardia (SB; 72.38%), and Complete Atrioventricular Block (CAVB; 45.35%) (Table 4).

In the reports full blood count values, renal and liver parameters, protein C and S activity, antithrombin III, factor VIII, homocysteine, vitamin B12, folic acid, fibrinogen protein, blood chemicals, chest X ray, hematology, sedimentation, blood biochemistry, neurological tests, cardiac enzymes, coronary

Table 1

The demographic and clinical findings of the patients diagnosed with mad honey poisoning (n = 1199).

Patient (P)-sex/age	Complaints	SBP-DBP mm Hg	HR beats/ min	ECG	Amount of honey consumed	Drug/disease	Treatment	Recovery time (h)	Ref no
1P-M/66	3, 8	40–70	35	SB, MI	—	—	2 mg atropine, SF	6	Akinci et al., 2008
1P-M/64	6, 7, 8, 10, 21	45–70	53	SB	—	None	1 mg atropine, SF	—	Aksel et al., 2014
1P-M/74	3, 8, 9	40–80	45	SB, STE	—	None	2 mg atropine	4–24 h	Alemdar et al., 2013
173P(91M–82F)	4, 8	—	—	SB, AVB	3–10 tbs	—	Pm	3–<12 h	Aliyev et al., 2009
1P-70/M	8, 9	60–80	—	CAVB, NR	—	None	SF	24	Altun et al., 2014
1P-M/79	2, 8, 10	30–50	38	SB, NR	—	None	0.5 mg atropine, SF	24 h	Alp et al., 2012
5P-5M	8, 12, 18	40–95	37–52	SB	—	None	1 mg atropine	12–24 h	Atik and Seliman, 2013
32–63	—	—	—	—	—	—	—	—	—
1P-M/36	5, 6, 8, 10, 14, 27, 31, 35	55–110	56	AVB, SB, NR, AF	2–3 tbs	—	SF	2 h	Bayram et al., 2012
16P (14M 2F) 41	1, 21	—	—	—	—	—	Atropine	—	Biberoğlu et al., 1988
1P-67/M	2, 6, 10, 11	40–60	40	SB	1 tbs	None	1 mg atropine	2	Bilir et al., 2014
1P-M/46	2, 8, 12, 19	50–80	40	SB	1 tbs	None	Atropine, SF	72 h	Binnetoglu et al., 2012
33P(30M–3F) (52)	2, 6, 8, 10, 12, 19, 28	46.42–77.86	55.35	SB	43.88 g	—	SF, 1–2 mg atropine	—	Bostan et al., 2010
1P-M/48	8, 9	60–80	33	—	100 ml	None	1 mg atropine	—	Caglı et al., 2009
1P-M/46	—	60–90	—	WPW, AF, QRST	—	None	SF	—	Cakar et al., 2012
1P-F/55	6, 7, 8	40–70	39	CAVB	50 ml	None	3 mg atropine, SF	6	Canpolat et al., 2012
2P-(1M–1F)/73–74	2, 6, 8, 9, 19, 5, 8	40–70	47	SB	Honey syrup	Hypertension	0.5 mg atropine, SF	6–24	Cavus et al., 2010
1P-M/43	3, 6, 10, 12	50–90	47	SB	—	Goiter	SF	—	—
1P-M/48	8	93–140	88	SB	—	Diabetes	SF	3–8 h	Cetin et al., 2009
1P-67/M	8, 9	36–74	37	AMI	30 ml	None	0.6 mg atropine	—	Chen et al., 2013
1P-1M/58	2, 6, 7, 8	50–70	45	JR, SB	20 ml	None	300 mg aspirin	—	—
21P(13M–8F)/(45–72)	1, 4, 6, 7, 8, 9, 10, 13	50–80	38	AVB, NR	150 g	None	Atropine, SF	7	Choo et al., 2008
37P (31M–6F)/56.17–69.66	1, 3, 6, 7, 8, 10, 11, 14, 21	78–125	56	SB, NR, AF	—	None	1 mg atropine	48	Cicek et al., 2004
21P (18M–3F) 41–86	1, 6, 7, 9, 10, 19, 21, 27	30–80	30–112 (56)	CAVB, SB, AF	1–6 tbs	Hypertension	1–2 mg atropine	—	Demir Akca and Kahveci, 2012
1P-1M/60	1	40–130	—	—	—	Diabetes	Dopamine, pm	—	—
1P-M/8	6, 8, 9, 10, 11	60–120	30–59	SB, AF, JR, LBB	50–300 g	Peptic ulcer	0.5–1 mg atropine	7–48 h	Demircan et al., 2009
1P-M/56	6, 8, 9, 12	45–85	45	SB	—	Parkinson's disease	Atropine	24	Desel and Neurath, 1998
1P-M/65	8,28,33	40–60	43	SB	30 ml	None	Dopamine	24	Dilber et al., 2002
1P-M/56	4	40–70	52	—	2 tbs	Hypertension	0.5 + 0.5 mg atropine, SF	24 h	Dubey et al., 2009
1P-M/28	2	80	30	CAVB	75 ml	Dyslipidemia	0.5 mg atropine, SF	—	—
1P-1M/52	10, 32, 39	40–70	52	—	2 tbs	Aspirin etc.	0.5 mg atropine, SF	—	—
1P-M/56	1, 4, 22	83–110	93	—	—	Diabetes	2 mg atropine, SF, pm	12	Dursunoğlu et al., 2007
38P (30M–8F)/22–79	3, 6, 8, 10, 11	51–89	48	SB	—	None	50 g active charcoal, SF	6 h	Eken, 2004
1P/42	8, 10, 14, 15, 19, 27, 35, 36, 37, 38	83–110	93	—	—	None	1 mg atropine	—	Engel et al., 2014
2P-1M–1F	1, 7	—	—	CAVB	—	None	1 mg atropine	—	Ergun et al., 2005
8P (6F–2M)/35–75 y	1, 6, 10, 21	—	31–47	—	—	None	—	—	Eroglu et al., 2013
1P-M/60	7, 8	60–100	36	—	—	Atropine	Atropine	—	Gerke et al., 2003
47P (40M–7F)/19–79	1, 8, 24, 25, 26	60–100	36	—	—	Metoclopramide, SF	—	24	Gossinger et al., 1983
30P(24M–6F)/	5, 6, 7	52.6–83.6	44.5	—	70 g	—	—	—	Gunduz et al., 1983
29P (25M–4F)/16–65	2, 6, 8, 10, 16, 18, 19	40–114	40–114 (53)	CAVB, SB, AV, NR	—	None	0.5 mg atropine, SF, pm	2–6	Gunduz et al., 2006
1P-M/66	6, 10, 18, 19	80–170	40	SB	—	None	0.5 mg atropine, pm	24 h	Gunduz et al., 2007
—	—	60–103	40	SB	—	None	Atropine, SF, pm	3.4–22	Gunduz et al., 2009
—	—	—	—	—	—	None	1 mg atropine, SF, pm	—	Gunduz et al., 2012
—	—	—	—	—	—	None	Ozone, treatment	—	Guven et al., 1989
—	—	—	—	—	—	None	Atropine, SF	—	—
—	—	—	—	—	—	None	Heparin	24 h	Halac et al., 2014

Table 1 (continued)

Patient (P)-sex/age	Complaints	SBP-DBP mm Hg	HR beats/ min	ECG	Amount of honey consumed	Drug/disease	Treatment	Recovery time (h)	Ref no
72P (59M–13F) 23–48	1, 21	40–130	38–58	CAVB	1–4 tbs	None	0.5–3 mg atropine, SF, pm	–	Hanci et al., 2010
1P-M/60	8, 18	20–90 50–70	(47.79) 60	SB AVB	2–3 tbs	None	SF	–	Icme and Cevik, 2010
1P-M/24	2, 6, 8, 10	45–95	50	SB	3–4 tbs	None	1 mg atropine, SF	24 h	Inal et al., 2013
1P-M/46	2, 6, 7, 8	50–80	28	SB	–	–	1 mg atropine	–	Inci et al., 2007
7P-7M/(20–25)	–	40–110	48–82	–	3–10 tbs	None	–	–	Jauhari et al., 2009
2P (1M–1F)	2, 5, 6, 9, 10	35–70	36, 38	SB, SB	1 tbs	None	–	–	Kaldırım et al., 2013
1P-M/53	6, 9, 10	40–70	30	SB, AF	–	None	0.5 mg atropine, SF	24 h	Kalkan et al., 2012
1P-M/29	6, 8, 10	50–90	42	SB	3–4 tbs	None	2 mg atropine	38 h	Kocak et al., 2008
1P-M/64	1, 3, 7, 10, 15, 19, 36, 37, 40	50–70	48	SB, AVB, BR	–	–	Acetylsalicylic acid Heparin Piritramide 0.5 mg atropine Metoclopramide	24 h	Krösche et al., 2007
1P-M/49	5	45–90	38	SB	–	–	–	–	Malottki and Wiechmann, 1996
1P-M/53	6, 9, 10	–	30	AF	Few tbs	None	0.5 mg atropine, SF	–	Memetoglu et al., 2011
2P	1, 7	–	–	SB, AVB	–	–	–	24 h	Oh et al., 2000
1P-F/45	5, 6, 8, 12, 17	50–70	44	AVB	2–3 tbs	None	1 mg atropine, SF	–	Oguzturk et al., 2012
42P-(33M 9F)	2, 6, 8, 10, 12	73.1–52.1	–	SB, CAVB, NR	–	–	Atropine, dopamine, SF	–	Okuyan et al., 2010
1P-M/54	3, 6, 8, 10	40–70	40	AF	2 tbs	None	1 mg atropine	–	Osken et al., 2012
19P (12M–7F)/22	6, 8, 9, 10, 12, 21	30–60 60–95	34–49	SB, AVB	30–130 g	Duodenal ulcer	0.5 mg atropine	24 h	Ozhan et al., 2004
2P (1M–1F)	6, 7, 8, 10, 20	60–80	48, 52	AVB	4 tbs	None	1 mg atropine, pm	–	Ozturk et al., 2011
55, 8	–	80–120	–	–	–	–	–	–	–
3P-3M (40–65)	2, 8, 9	50–80	39–59	SB	1tbs	None	1 mg atropine, SF	–	Ozturk et al., 2014
1P-M/35	8, 19	35–70	40	SB	1 tbs	Asthma	0.5 mg atropine, SF	24 h	Sahin et al., 2013
1P-M/55	6, 9, 10, 11	40–70	32	SB	–	None	0.5 mg atropine, SF	24 h	Sahpaz et al., 2014
1P-M/70	3, 6, 7, 8, 10, 17, 19	20–60	105	BB	Few tbs	–	SF	6 h	Saritas et al., 2011
1P-F/87	7	30–60	48	SB	–	None	Dopamine, SF	48 h	Sayin et al., 2011
1P-M/76	3, 6, 8, 10	40–60	30	AVB, LBB, ST-SE	3 tbs	Hypertension Lung disease Pneumonia	1 mg atropine	–	Sayin et al., 2012
3P	1, 21	–	–	SB	–	–	–	3 days	Sogut et al., 2009
1P-M/42	8, 10, 12, 18, 19	50–70	44	SB	2–3 tbs	None	1 + 1 mg atropine	24 h	Sohn et al., 2005
3P	1, 7, 21	–	–	SB	–	–	atropine, SF	–	Sumerkan et al., 2012a
1P-M/29	8, 9	50–80	39	CAVB	1 tbs	None	1 mg atropine	–	Sumerkan et al., 2012b
116P-(87M 29F)/50	4, 21	40–80	46	AVB, SB, STSE, STD	21.25 g	None	Atropine, SF	4–24 h	Sumerkan et al., 2013a
246P(188M–58F)/ 55.5	7, 21	40–80	46	SB, STE, CAVB	21.25 g	None	Atropine, SF	3.5–24 h	Sumerkan et al., 2013b
1P-M/54	7	–	32	AVB	4–5 tbs	None	1 mg atropine	24 h	Tulmac et al., 2008
1P-M/15	8, 34	50–75	45	SR	150 ml	None	0.5 mg atropine, SF	–	Uzun et al., 2013a
46P-(36M–10F)/ 52.2	–	37.8–70.8	44.7	CAVB	–	None	–	–	Uzun et al., 2013b
6P-6M/20–22	1, 6, 19, 21	–	–	SB	–	None	0.5–1 mg atropine	–	Unlu et al., 2010
7P–1F/45–6M (22 –56)	1, 2, 6, 8, 10, 11, 12, 21, 27, 29	30–87 60–142	28–94	–	15–30 g	–	Atropine, SF	–	Weber et al., 2009
1P-M/70	6, 7, 10, 19	60–110	34	AVB, SDHB	–	Metformin/ diabetes	Atropine	72 h	Weiss et al., 2010
2P(1M–1F)/42, 50	3, 20	40–85	35–45	CAVB, NR, STSE	1 tbs	–	1 mg atropine	6 h	Yarlioglu et al., 2011
23P-(21M–2F)/7 months–61 y	2, 6, 8, 10, 11, 12, 16, 18, 19, 21, 22, 23	–	–	CAVB, SB, JR	2–5 tbs	–	Atropine, SF, isoproterenol	0.5–6 h	Yavuz et al., 1991
16P (10M–6F)/41 –79 (58.5)	4, 7, 8, 10, 19, 11	45–73	42	WPW SB, NR, AVB AF	2.9 g	None	1.1 mg atropine	27.7 + 7.2 h	Yaylacı et al., 2014
1P-F/63	3, 6, 8, 12	40–60	40	SB	2 tbs	None	1 mg atropine, SF	16 h	Yaylacı et al., 2011
1P-58M	6, 8, 10, 19	30–50	41	SB	–	None	2 mg atropine	24 h	Yaylacı et al., 2013
4P(1M–3F)/5–38	2, 6, 8, 10, 15, 18, 19	30–90	42–66	SB	1–2 tbs	None	1 mg atropine, SF	–	Yengil et al., 2013
1P-M/41	3	30–60	96	MI	–	None	aspirin, SF, heparin	2 h	Yıldırım et al., 2008

(continued on next page)

Table 1 (continued)

Patient (P)-sex/age	Complaints	SBP-DBP mm Hg	HR beats/ min	ECG	Amount of honey consumed	Drug/disease	Treatment	Recovery time (h)	Ref no
66P-(52M–14F)/18–85 y	8, 9, 19, 6, 10, 7, 14	50–100 30–60	27–66 (47.96)	—	5–30 g	None	0.5–2 mg atropine, SF	24 h	Yılmaz et al., 2006
1P-M/50	3, 30, 32	55–80	44	JR	2 blows	None	3 mg atropine, SF	12 h	Yorgun et al., 2008

M, male patient; F, female patient; SBP, systolic blood pressure; DBP, diastolic blood pressure; SF, saline solution for intravenous infusion; pm, pace maker; tbs: tablespoon; —: numeric data not available.

SB, sinus bradycardia; CAVB, complete atrioventricular block; AVB, atrioventricular block; ST-SE, ST-segment elevation (ST-SE); NR, nodal rhythm; AF, atrial fibrillation; JR, junctional rhythm; MI, myocardial infarction; LBB, left bundle branch block; WPWS, Wolf Parkinson White Syndrome, QRST complex, QRST waves; BR, bradyarrhythmia; SDHB, second degree heart block.

Complaints: 1.Hypotension 2. Fainting 3.Chest pain 4.Presyncope 5. Vertigo/headache 6. Nausea 7. Syncope 8. Dizziness 9. Weakness 10.Vomiting 11. Impaired consciousness 12. Sweating 13. Excessive perspiration 14. Salivation 15. Cramp 16. Chills 17. Gastroenteritis 18. Exhausting 19. Blurred vision 20. Malaria like fever seizure 21. Bradycardia 22. Collapse 23. Cyanosis 24. Ataxia 25. Bradyarrhythmia 26. Diaphoresis 27. Parasthesia 28. Mental confusion 29. Diarrhea 30. Palpitation 31. Diplopia 32. Dyspnea 33. Cold sweating 34. Malaise 35. Lightheadedness 36. Agitation 37. Colic 38. Coma 39. Dysphagia 40. Tachycardia.

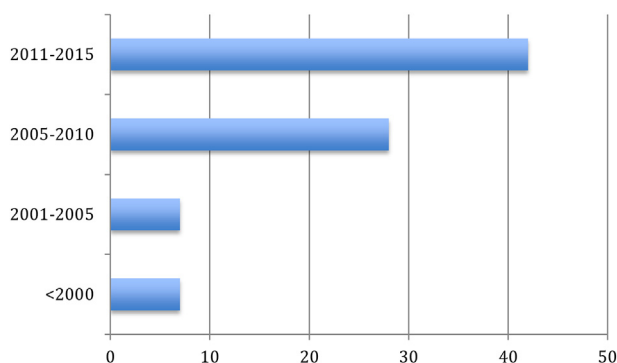


Fig. 1. Cases on mad honey intoxication between <2000 and 2015 years.

angiogram, troponin levels, CBC and serum biochemistry, physical examination, kidney and liver function tests, electrolytes, total CK and CK-MB, PT-PTT and urine biochemistry, ALT AST, creatine kinase, thyroid function tests, motor exam, telecardiography results were investigated and found to be normal. Moreover, 75.44% of patients were on some medication or had a chronic disease such as hypertension, type 2 diabetes, gastritis, dyslipidemia, lung disease, hepatitis. The most of the medication used were metformin, aspirin, nitrates, atenolol, atorvastatin, tacrolimus, ianovidin and antihyperglycemia.

A list of adapted treatments for the patients in 1199 cases and the dose used in these efforts can be also provided in Table 5. The most common agents used for the treatment were atropine and SF at different doses. However, 20.74% of patients were treated with other agents such as aspirin, heparin, isoproterenol, enoxaparin, charcoal (50 mg), acetylsalicylic acid (ASA), piritramide,

metaclopramide. Furthermore, 14.90% of patients were admitted to the intensive care unit. The patients reviewed in this study were treated in hospital for different durations. According to data available in only a fraction of (49) cases, majority (71.55%) of patients were discharged within 12 h, while 28.45% of the patients remained hospitalized for more than 24 h (Table 5).

In 28.3% of cases the amount of the honey consumed was not known. But, 75.12% of the patients defined the amount as 1–5 tablespoon (Fig. 2). In addition to this amounts as 2 blows and 100 ml/day/a week or usage as honey caudle have been reported in the literature.

4. Discussion

Historically, the phenomenon of honey intoxication is known from ancient times. Xenophon (B.C. 434–354) mentions an event of intoxication of 10,000 soldiers during their stay at the coast of the Black Sea with the Greek Army (Kelhoffer JA, 2005). However, the accessible case reports on mad honey intoxication started in 1983 (Gossinger et al., 1983).

Previous studies have reported that mad honey intoxication is more common among middle-aged males (Yılmaz et al., 2006; Uzun et al., 2013a,b; Hancı et al., 2010). In consistent with results of our review, mad honey intoxication occurred more often in males compared to females (Uzun et al., 2013a,b; Hancı et al., 2010). It may be related to the incidence of high blood pressure and sexual dysfunction, more common among middle-aged males. Hypertension is observed more frequently in males and its incidence increases with age (Carretero and Oparil, 2000.) As mad honey is frequently consumed in traditional medicine, especially for the treatment of high blood pressure, constitutes the largest age group of patients affected. Besides, it has also been reported that mad

Table 2

Patient characteristics, diastolic blood pressure (DBP), systolic blood pressure (SBP) and pulse rate.

Patient characteristics	n	%	Minimum	Maximum	Mean ± SD
Age					
<1 years	1	0.08			
1–15 years	4	0.34			
15–40 years	83	6.97			
41–65 years	990	83.19			
>65 years	112	9.41			
Age, years (mean ± SD)			7 months	87 years	52.84 ± 18.08
Sex					
Male	896	75.17			
Female	296	24.83			
DBP mmHg (mean ± SD)			20	93	46.99 ± 14.41
SBP mmHg (mean ± SD)			50	170	86.02 ± 24.81
Pulse rate, beats/min (mean ± SD)			28	105	47.11 ± 15.87

Table 3
Complaints of patients.

Complaints	n	%	Complaints	n	%
1. Hypotension	237	19.75	21. Bradycardia	599	49.92
2. Fainting	176	14.66	22. Collapse	24	2.00
3. Chest pain	110	9.16	23. Cyanosis	23	1.92
4. Presyncope	328	27.33	24. Ataxia	47	3.92
5. Vertigo/headache	37	3.08	25. Bradyarrhythmias	46	3.83
6. Nausea	451	37.58	26. Diaphoresis	47	3.92
7. Syncope	192	16.0	27. Paraesthesia	30	2.50
8. Dizziness	619	51.58	28. Mental confusion	34	2.83
9. Weakness	146	12.17	29. Diarrhea	8	0.67
10. Vomiting	426	35.50	30. Palpitation	1	—
11. Impaired consciousness	140	11.67	31. Diplopia	1	—
12. Sweating	136	11.33	32. Dyspnea	1	—
13. Excessive perspiration	22	1.83	33. Cold sweating	1	—
14. Salivation	105	9.04	34. Malaise	1	—
15. Cramp	4	0.34	35. Lightheadedness	1	—
16. Chills	52	4.48	36. Agitation	2	—
17. Gastroenteritis	2	0.17	37. Colic	2	—
18. Exhausting	64	5.33	38. Coma	1	—
19. Blurred vision	245	20.42	39. Dysphagia	1	—
20. Malaria-like hyperthermia	4	0.33	40. Tachycardia	1	—

Table 4
ECG characteristics of patients.

ECG characteristics	%
Sinus bradycardia (SB)	79.58
Complete atrioventricular block (CAVB)	45.83
ST-segment elevation (ST-SE)	22.63
Atrioventricular block (AVB)	30.91
Nodal rhythm (NR)	11.27
Atrial fibrillation (AF)	8.69
Junctional rhythm (JR)	4.04
Myocardial infarction (MI)	0.26
Left bundle branch block (LBB)	0.09
Wolf Parkinson White Syndrome (WPWS)	0.09
QRST complex	0.09
Asystole	0.09
BR	0.09

honey is used by between 40 and 60 years old men most commonly to cure sexual dysfunction and increase sexual performance (Demircan et al., 2009; Demir Akca and Kahveci, 2012). Data from the Massachusetts Male Aging Study (MMAS) shows that 34.8% of men aged 40–70 years had moderate to complete erectile dysfunction (Feldman et al., 1994). Our data also lends support to this correlation and the male population that is more frequently affected by honey poisoning shows concordance with the age group suffering from sexual dysfunction.

Most common signs of mad honey intoxication are “bradycardia” and “hypotension” which can cause the symptoms such as dizziness, nausea, vomiting, weakness, blurred vision, syncope, vertigo/headache, impaired consciousness, sweating, fainting, hypotension, malaise, excessive perspiration, chest pain, presyncope,

exhausting (Özhan et al., 2004). In other words most of the symptoms are related to these two basic signs. Bradycardia in an adult is any heart rate less than 60 beats per minute (BPM), although symptoms usually manifest only for heart rates less than 50 (Neumar et al., 2010). Symptoms of bradycardia include weakness, easy fatigability, lightheadedness, dizziness, and loss of consciousness (Mangrum and DiMarco, 2000). Lower blood pressure can be a side effect of certain botanicals (Tabassum and Ahmad, 2011; Mitchell et al., 2011) and in indigenous medicine they have been used to treat high blood pressure (Kelly, 2005). However, *Rhododendron* is unique for being used in the honey form.

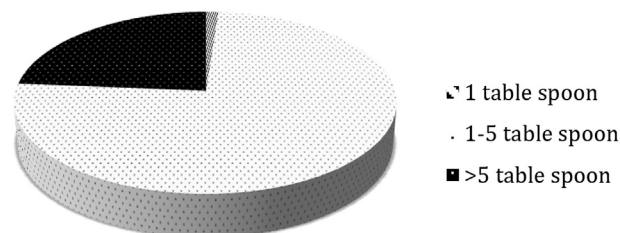
Hypotension sufficient to cause cerebral symptoms can develop insidiously over minutes. Although symptoms of faintness and giddiness predominate, some patients lack such prodromal warnings, presumably because of the absence of strong efferent parasympathetic activity. Sometimes chronically ill patients become confused or tremulous without the usual sensations of faintness or collapse. Diagnosis comes from observing an acute decline in the mean blood pressure of more than 10–15 mmHg and/or in the heart beat rate more than 60 bpm (Frederique et al., 2001).

In the previous studies it has been reported that consuming various quantities of mad honey (5–180 g) may cause intoxication (Gunduz et al., 2006, 2009; Ozhan et al., 2004). However, some researchers suggested that in mad honey intoxication cases, rather than the amount of honey intake the exact amount of GTX is important (Demir Akca and Kahveci, 2012). As a matter of fact, the distribution of GTX within honey is not homogeneous and it is not easy to determine the exact amount of GTX exposure (Gunduz et al., 2006; Aliyev et al., 2009).

The honeys produced in regions where *R. ponticum* and *R. luteum* plants are common, interfere with nectars of other plants,

Table 5
Treatment of patients.

Treatment	%
Atropine	79.26
0.5 mg atropine	37.79
1 mg atropine	49.73
2 mg atropine	8.03
3 mg atropine	4.45
Saline infusion (i.v.)	65.35
Others (dopamine, aspirin, heparin, dopamine, isoproterenol, active charcoal, ozone treatment, metoclopramide, piritramide)	20.74

**Fig. 2.** Amount of consumed honey of patients.

generally the chestnut and lime, which blossom in the same season (Silici et al., 2014). Therefore, it is not possible to determine that this honey has been produced solely from rhododendron nectar. Thus, in the presence of intoxication, the amount of honey consumed cannot be used as criteria.

Atropine inhibits the muscarinic actions of acetylcholine at postganglionic parasympathetic neuroeffector sites including smooth muscle, secretory glands and central nervous system (CNS) sites. It is particularly useful in relieving bronchoconstriction and salivation induced by anticholinesterases (Heath and Meredith, 1992). Specific anticholinergic responses are dose-related. In this review moderate dose (0.5 and 1 mg) of atropine appears to be effective. Kentala et al. (1990) reported that moderate doses increased the heart rate (vagolytic effect). Large doses may block nicotinic receptors at the autonomic ganglia and at the neuromuscular junction. Large doses of atropine impair accommodation, causing dilation of the pupil and blurred vision (Weiner, 1985). When given alone atropine has little effect on blood pressure, although it can block completely the hypotensive and vasodilatory effects of choline esters. In addition, small doses of atropine have little depressant action on the CNS. However, in toxic doses, atropine initially causes central excitation (exhibited as restlessness, confusion, hallucinations, and delirium) followed by central depression with coma and death. Atropine is incapacitating at doses of 10–20 mg per person. Its LD50 is estimated to be 453 mg per person (per oral) with a probit slope of 1.8 (Goodman E, 2010).

A careful history and physical examination, including any possible information gained from witnesses, generally give more diagnostic information than any other approach. Such patients rarely require an evaluation more elaborate than a careful history and physical examination plus ECG, standard blood counts and blood chemistry determinations.

5. Conclusion

1199 cases were evaluated and the following results were obtained: a) Mad honey is consumed more by middle age men, b) main complaints of those who consume mad honey are dizziness, bradycardia, nausea, vomiting and presyncope, c) The most important parameters in clinical diagnosis are low pulse rates and drop in blood pressure. After determining that mad honey was consumed by a patient, further tests could not be needed, d) consuming 1–5 table spoons of mad honey usually causes intoxication, e) treatment of 0.5–1 mg atropine and salin i.v. is generally effective, f) If not really needed, patients should not be taken into intense care unit and be discharged within 24 h.

Although the phenomenon of honey intoxication is known from ancient times the accessible case reports have started recently. Since rather than the honey itself the amount of GTX is of importance in the presence of intoxication, the amount of honey consumed cannot be used as criteria. It is important that patients presenting to Emergency Department with bradycardia and hypotension should be considered for intoxications in a wide spectrum, with the consideration of local and non-local foods as a causative agent. To establish the diagnosis of mad honey intoxication, a clinical evaluation that includes a detailed history is generally sufficient. Medicinal use of Rhododendron honey has not been well understood yet, therefore care should be taken when consuming GTX containing honey.

Conflicts of interest

The authors declare that there are no conflicts of interest.

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